

Simulation Results Explanation

This document provides a detailed explanation of the physical interpretations and core conclusions for the five sets of visualization charts generated by the `simulation.py` code. The simulation aims to verify the effectiveness of the **intent-driven multi-timescale collaborative optimization mechanism for open infrastructure**.

1. Real-time Throughput & SLA Violation

(Corresponds to `visualization 1` in the code)

Chart Description

- **Top Chart (Real-time Throughput):** Displays the real-time throughput curves for all active slices during the simulation period.
 - **Solid Line:** Actual achieved throughput.
 - **Dashed Line:** The minimum throughput baseline required by the slice SLA.
 - **X-Axis:** Time Steps (1 Step = 1 PS cycle; 10 Steps = 1 PM cycle).
- **Bottom Chart (SLA Violation Magnitude):** Displays the magnitude of SLA violations (Mbps) for each slice.

Core Conclusions

1. **Attribution of Dynamic Fluctuations:** The throughput curves exhibit dynamic fluctuations, primarily resulting from the coupling of **radio channel randomness** (MCS varying between 5-28) and the **compute resource following mechanism** (the PM layer dynamically adjusts CPU based on the radio potential of the previous cycle).
2. **Staged Access Performance:**
 - **Step 0-60 (Resource Sufficient Period):** Despite channel fluctuations, the dynamic algorithm consistently maintains throughput above the SLA baseline, achieving robust guarantees.
 - **Step 60+ (Saturation Period):** The access of all slices leads to system resource saturation. Minor violations occur due to physical bottlenecks, which aligns with physical laws.

2. Optimized Deployment Topology

(Corresponds to `visualization 2` in the code)

Chart Description

- Visualizes the physical deployment decisions made by the Long-term Planning Layer ($\mathcal{P}_L$) for different slice network functions (APP, UPF, CU-UP).
- **Zone Division:** Edge DC, Regional DC, Central DC.

Core Conclusions

- **SLA-Driven Intelligent Deployment:** The algorithm successfully translates high-level intents into physical topological constraints.
 - **Latency-Sensitive Slices (e.g., S4, S5 URLLC):** Automatically deployed at the **Edge DC** to meet strict latency constraints (<10ms).
 - **Cost-Sensitive/Latency-Tolerant Slices (e.g., S1, S2 mMTC):** Consolidated at the **Central DC** to minimize infrastructure rental costs.
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3. Bottleneck Analysis

(Corresponds to `visualization 3` in the code)

Chart Description

- Provides a microscopic view of resource coordination for a single slice across different time scales.
- **Blue Line (Radio Guaranteed):** Represents the **guaranteed radio bandwidth** (i.e., $b_{min} \times \eta$) decided by the PS layer optimization model.
- **Green Line (Compute Cap):** The upper limit of computing capacity allocated by the PM layer (Medium-term).
- **Red Line (Actual Throughput):** The actual throughput achieved.

Core Conclusions

- **Radio Side Dynamic Adjustment (b_{min} Decision):** The fluctuation of the blue line intuitively demonstrates the effectiveness of the PS layer algorithm. Under continuous random fluctuations of radio channel states (MCS/ η), the algorithm dynamically adjusts the number of reserved resource blocks (b_{min}) in real-time to ensure that the guaranteed bandwidth provided by the radio side (Blue Line) never falls below the minimum SLA throughput requirement.
 - **Compute Side Synergy:** The Green Line (Compute Capacity) dynamically tracks the Red Line (Actual Demand), preventing compute bottlenecks and demonstrating the cross-layer coordination mechanism.
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4. Resource Utilization

(Corresponds to `visualization 4` in the code)

Chart Description

- Displays the CPU and RAM resource utilization over time for Edge, Regional, and Central data centers.

Core Conclusions

- **Validation of Parameter Rationality:** The resource utilization of Edge and Regional nodes is very high (near saturation) during peak traffic periods. This strongly proves the **rationality of the simulation parameter settings**—we successfully constructed a realistic network scenario that is resource-constrained and highly competitive, rather than an ideal environment with infinite resources. This validates the necessity of optimization scheduling under limited resources.

5. Comparison: Dynamic vs. Static

(Corresponds to `Visualization 5` in the code)

Chart Description

- Compares the performance of the proposed **Dynamic Coordination Algorithm (Dynamic)** against a traditional **Static Baseline Policy (Static)**.
- **Solid Line:** Results from the Dynamic Algorithm.
- **Dashed Line:** Results from the Static Policy (One-time long-term planning only, without medium/short-term adjustments).

Core Conclusions

- **Vulnerability of Static Policy:** The static scheme cannot adapt to time-varying channel characteristics. When channel quality fluctuates continuously (e.g., during **Step 30-60** where slice access increases and channels vary), the fixed resources allocated statically cannot offset the fading effects, leading to severe SLA violations.
- **Robustness of Dynamic Algorithm:** The proposed algorithm possesses **cross-layer real-time compensation capabilities**. Under identical channel conditions, the algorithm actively increases radio resources (PS Layer) and coordinates compute expansion (PM Layer) to effectively offset the impact of channel fading, successfully maintaining minimum throughput and significantly reducing the risk of violations.